

Total No. of Questions : 10]

SEAT No. :

P230

[Total No. of Pages : 2

[5871]-757

B.E. (Honors)

**APPLICATION DEVELOPMENT USING AUGMENTED
REALITY AND VIRTUAL REALITY
(2015 Pattern) (Semester - II) (410703)**

Time : 2½ Hours]

[Max. Marks : 70

Instructions to the candidates:

- 1) Solve Q.1 or Q.2, Q.3 or Q.4, Q.5 or Q.6, Q.7 or Q.8, Q.9 or Q.10.
- 2) Neat diagrams must be drawn whenever necessary.

Q1) a) Differentiate between Virtual reality and Augmented Reality and Mixed reality? [5]

b) What are the software and hardware requirement of VR integration? [5]

OR

Q2) a) Explain the applications areas of AR and VR? [5]

b) What is Google VR SDK? [5]

Q3) a) Explain text optimization in VR. [5]

b) Explain anatomy of SLAM. [5]

OR

Q4) a) What are different features of AR toolkit? [5]

b) How to do Loop detection and loop closing in AR? [5]

Q5) a) Explain different classes in C#? [6]

b) Explain functions and macros in C# unit with suitable example? [12]

OR

Q6) a) How to define variables in C# Unity? Explain with example. [6]

b) Explain dynamic memory allocation with suitable example in C# with Unity. [12]

P.T.O.

- Q7)** a) Explain advantage and disadvantage of AR and VR technologies. [5]
b) What are different AR components? [3]
c) Explain structure and working of HTC Vive. [8]

OR

- Q8)** a) What is Google Cardboard? [4]
b) Explain structure and working of Samsung gear VR. [6]
c) Explain Virtual Retinal System in detail. [6]

- Q9)** a) Explain different human factors considered for AR VR? [6]
b) Write the use cases for AR VR in health and medicine. [10]

OR

- Q10)** a) Explain legal and social consideration factors of AR VR. [6]
b) Explain with example use of AR VR in Telerobotics and Telepresence. [10]
